



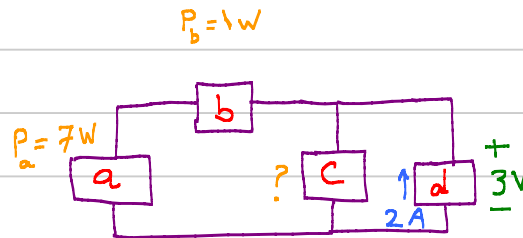
Ex. The power associated with element "a" and "b" are given. Find the power of element "c".

$$P_d = -vi = -(3)(2) = -6W$$

$$\sum P = 0$$

$$7 + 1 - 6 + P_c = 0$$

$$P_c = \underline{-2} W$$



### Five Basic Circuit Elements:

- voltage source
- current source
- resistor
- inductor
- capacitor

} active circuit elements  
(can generate electric energy)

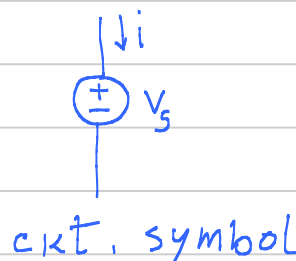
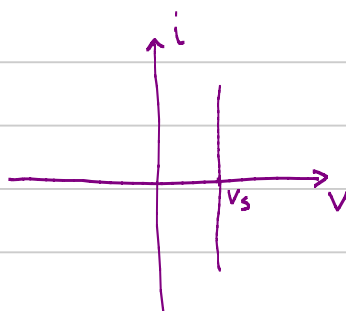
} passive circuit elements  
(cannot generate electric energy)

### Ideal Sources

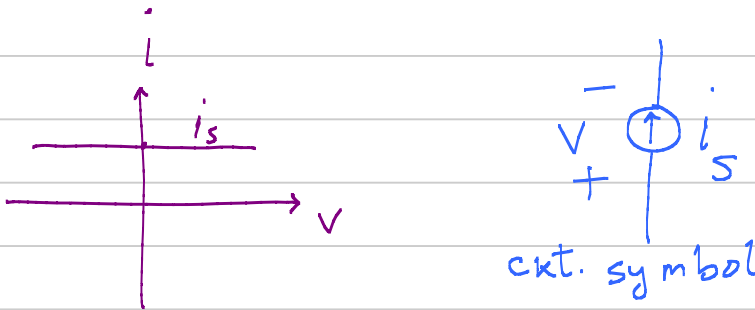
There are two types of ideal sources: dependent sources and independent sources.

An independent source provides a specified voltage or current independent of the other circuit elements connected to it.

**independent voltage source:** maintains the specified voltage regardless of the current passing through it.



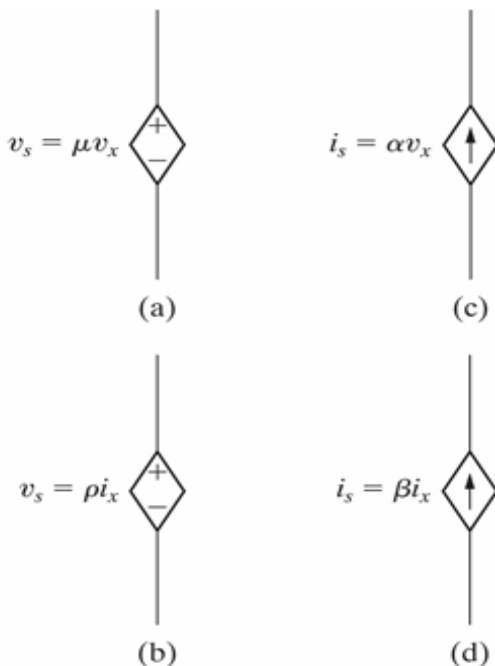
**Independent current source:** maintains the specified current regardless of the voltage across it.



The relationship between the voltage and current of an independent source is determined by the other elements connected to it.

### Dependent Sources

The value of a dependent source is controlled by another voltage or current in the circuit. There are 4 types of dependent sources:



a) voltage controlled voltage source

b) current controlled voltage source

c) voltage controlled current source

d) current controlled current source

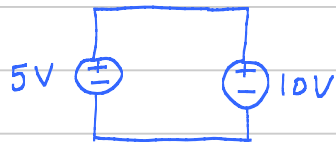
$\mu, \rho, \alpha$  and  $\beta$  are constant.

$\mu$ : dimensionless

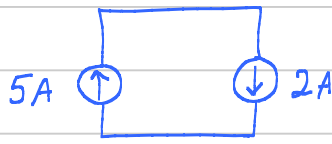
$\rho$ :  $\frac{V}{A} (\Omega)$ ,  $\alpha$ :  $\frac{A}{V} (S)$

$\beta$ : dimensionless

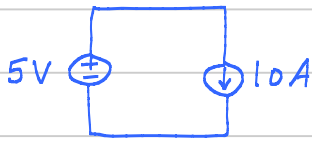
Ex. Which interconnection is valid?



Not valid



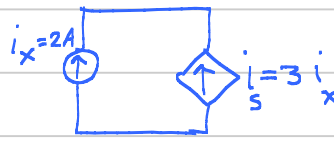
Not valid



✓



✓



Not valid

Ex. Compute the power associated with each element.

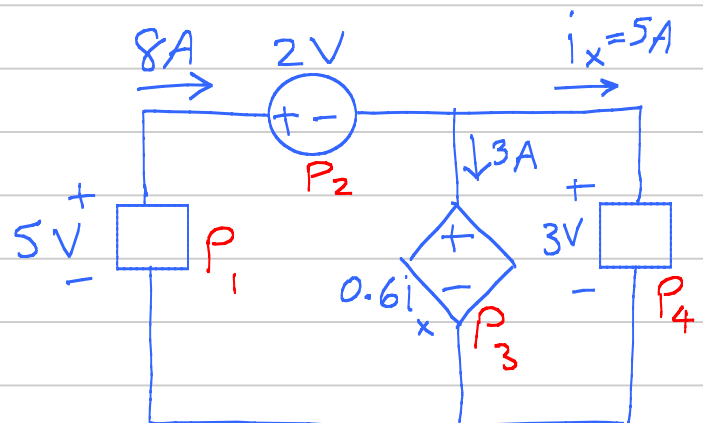
$$P_1 = -vi = -(5)(8) = -40\text{W}$$

$$P_2 = vi = 2 \times 8 = 16\text{W}$$

$$P_3 = vi = (0.6 \times 5)(3) = 9\text{W}$$

$$P_4 = vi = (3)(5) = 15\text{W}$$

$$\text{check: } \sum P \stackrel{?}{=} 0$$



$$-40 + 16 + 9 + 15 = 0 \checkmark$$

## Resistor

Resistance is the capacity of material to resist the flow of electric current. It is denoted by  $R$  and measured in ohms.

$$R = \rho \frac{l}{A}$$



Resistor is a circuit element used to model the current resistivity of the resistor.

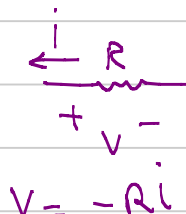
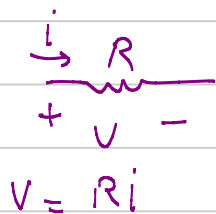


Basic Laws: Ohm's law, Kirchhoff's current law (KCL), Kirchhoff's voltage law (KVL).

**Ohm's Law:** The ratio of voltage to current of a resistor is constant and equals the resistance of the resistor.

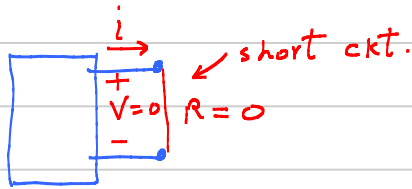
$$\frac{V}{i} = R \Rightarrow \boxed{V = Ri}$$

implementing passive sign convention:

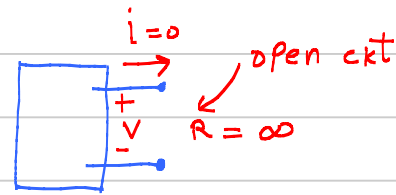


$R$  is always positive  $0 < R < \infty$

$R = 0$  (short ckt.)



$R = \infty$  (open ckt.)



current is determined by  
the ckt. inside the box

voltage is determined by  
the ckt. inside the box

Conductance is the reciprocal of resistance, denoted by  $G$  and measured in Siemens (S).

$$G = \frac{1}{R}, \quad i = GV \quad (\text{ohm's law})$$

Power for resistors

$$P = Vi = Ri \cdot i = Ri^2$$

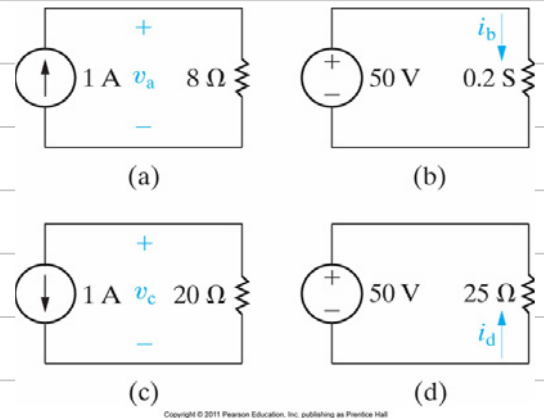
$$P = Vi = V \cdot \frac{V}{R} = \frac{V^2}{R}$$

$$P = Vi = i^2 R = \frac{V^2}{R}$$

The power of a resistor is always positive and  
it can only absorb power.

$$\text{OR, } P = \frac{i^2}{G} = V^2 G \quad (\text{in terms of conductance})$$

**Ex.** Calculate the unknown voltage or current and determine the power dissipated in each resistor.



$$a) V_a = 8 \times 1 = 8 \text{ V} , P = Vi = 8 \times 1 = 8 \text{ W}$$

$$b) i_b = VG = (50)(0.2) = 10 \text{ A}$$

$$P = V^2 G = (50)^2 (0.2) = 500 \text{ W}$$

$$(or P = Vi = 50 \times 10 = 500 \text{ W})$$

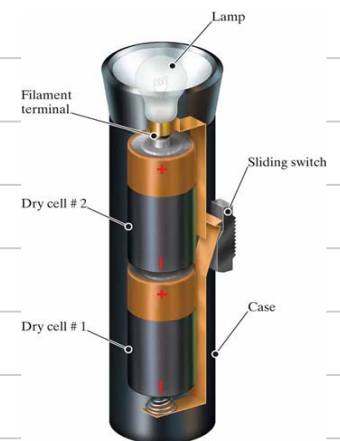
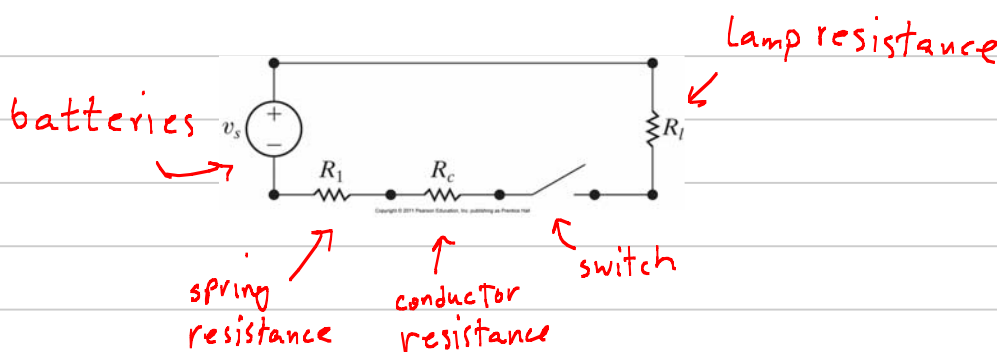
$$c) V_c = (-1)(20) = -20 \text{ V} , P_{20\Omega} = Ri^2 = (20)(-1)^2 = 20 \text{ W}$$

$$d) i_d = -\frac{50}{25} = -2 \text{ A} , P_{25\Omega} = Ri^2 = 25(-2)^2 = 100 \text{ W}$$

## Electric Circuit Models

The circuit elements are used to construct the circuit model for practical systems.

**Example:** circuit model of a flashlight



## Analogy of fluid flow system to electric circuits:

Voltage source ==> pump

Conductor ==> pipe

Current ==> flow rate

Voltage ==> pressure difference

Switch ==> valve

resistance ==> constriction